

Proof of Proposition 6. Both parts (a) and (c) follow exactly from the proof of Proposition 5. This is immediate from observing that the bound established in the proofs of Proposition 5(a) and Proposition 5(c) only consider the first potential sharer, who in this case is just the regulator (instead of the seeded agent). A similar example to that in Proposition 5(b) can be supplied by the authors. ■

Proof of Proposition 7. For both parts (a) and (c), consider a new (stricter) performance target $\lambda^* < \lambda$ (where λ is the current performance target). The platform can meet the new performance standard λ^* by recommending any content with $r \geq \phi^{-1}(\lambda^*)$, which by continuity of ϕ and positive density of f over $r \in [\phi^{-1}(\lambda^*), 1]$ guarantees some level of expected engagement $E > 0$ whereas (platform-optimal) engagement while neglecting the performance target is some $\tilde{E} \geq E$. Setting $\alpha^* = (\tilde{E} - E)/\tilde{E}$ implies that for all $\alpha > \alpha^*$, the platform prefers to meet λ^* than neglect it. Finally, note that to meet a lower performance target λ^* , the total shares on the platform (in expectation) must decrease.²⁷ Given that the performance target is the ratio of misinformation shares to total shares, and total shares have decreased, the platform must decrease misinformation shares to meet λ^* .

For part (b), we consider a complete sharing network with uniformly distributed beliefs on $[0, 1]$, but otherwise we adopt the same parameter values as in Example 1. Moreover, the auditing technology to enforce performance targets is only $\alpha = 1/2$. Suppose that there are just three types of articles: high-reliability and low-sensationalism (type HL) articles, low-reliability and low-sensationalism articles (type LL) articles, and low-reliability but high-sensationalism (type LH) articles. The type-HL articles never contain misinformation ($\phi(r) = 1$) and have no sensationalism ($\kappa = 0$). The type-LL articles contain misinformation with probability $2/3$ and have no sensationalism ($\kappa = 0$). The type-LH articles also contain misinformation with probability $2/3$, but have high sensationalism given by $\kappa > (2c - u)/(3N) > 0$. We assume that all three article types are equally likely. In the case of type-HL and type-LH articles, all agents necessarily share in the most-sharing equilibrium. On the other hand, only agents with priors $b_i \in [3/4, 1]$ share the type-LL articles. It is straightforward to verify that when $\lambda \in (34/99, 1) \cup (0, 1/3)$, the platform allows all content, but when $\alpha \in (1/3, 34/99)$, the platform removes the type-LL articles and reduces the virality of misinformation. ■

Proof of Proposition 8. The network regulation does not bind for an article with $r > r_P$, so we need only consider $r < r_P$. Take some agent i with prior $b_i \in (\bar{b}, \bar{b} + \eta)$ in a small neighborhood $\eta > 0$ of $\bar{b}$ (where $\bar{b}$ is the same $\bar{b}$ constructed in Theorem 3). Following the same line of reasoning as in Theorem 2(a), agents with priors in this interval elect to ignore instead of share following the network regulation (and when η is sufficiently small), and this necessarily reduces the virality of misinformation, showing (ii). To prove (i), we note that agents in this neighborhood around $\bar{b}$ also do not share in the most-sharing equilibrium under *any* sharing network P' (following the network regulation), per the construction of $\bar{b}$ in Theorem 3. Therefore, the platform cannot generate additional engagement by departing from the class of island models (specifically, two-island models) while maintaining $p_s/p_d \leq p^*$. ■

²⁷For part (c), this follows from noting that with the more relaxed $\lambda > \lambda^*$ target, the platform could have increased engagement with a different sharing network (and thus is not implementing a profit-maximizing sharing network).